$R(t) > 0$ love

$R(t) = 0$ indifferent

$J(t) < 0$ hates

$$\frac{d}{dt}\begin{pmatrix} R(t) \\ J(t) \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\begin{pmatrix} R(t) \\ J(t) \end{pmatrix}$$

a_{11} R responce to his own feelings

a_{22} J her

a_{12} R to J creating him

a_{21} J R her

Optimated Romeo

$$a_{12} = a_{11} = 0$$

$$\frac{d}{dt} R(t) = 0, \quad R(t) = R_0$$

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① $\frac{d}{dt} J(t) = a_{21} R(t) + a_{22} J(t)$

② $\frac{d}{dt} J(t) - a_{22} J(t) = a_{21} R_0$

③ $e^{-a_{22} t}$

④ $e^{-a_{22} t} \frac{d}{dt} J(t) - a_{22} e^{-a_{22} t} J(t)$

$= a_{21} R_0 e^{-a_{22} t}$

⑤ $\int_0^t \frac{d}{dt} \left(e^{-a_{22} t} J(t) \right) dt = \int a_{21} R_0 e^{-a_{22} t} \, dt$

⑥ $e^{-a_{22} t} J(t) - J(0) = a_{21} R_0 \frac{e^{-a_{22} t} - 1}{-a_{22}}$

⑦ $J(t) = J(0) e^{a_{22} t} - a_{21} R_0 \frac{1 - e^{a_{22} t}}{a_{22}}$

$$y'(x) + p(x)\,y(x) = q(x)$$

$$\mu(x) = e^{\int_0^x p(x')\,dx'}$$

$$e^{\int p(x)\,dx}\,y'(x) + p(x)\,e^{\int p(x)\,dx}\,y = q(x)\,e^{\int p(x)\,dx}$$

$$\frac{d}{dx}\left(e^{\int p(x)\,dx}\,y(x)\right) = q(x)\,e^{\int p(x)\,dx}$$

$$e^{\int p(x)\,dx}\,y(x) = \int dx\; q(x)\,e^{\int p(x)\,dx}$$

4 Do opposite attract?

$$\frac{d}{dt}\begin{pmatrix} R(t) \\ J(t) \end{pmatrix} = \begin{pmatrix} a & b \\ -b & -a \end{pmatrix}\begin{pmatrix} R(t) \\ J(t) \end{pmatrix}$$

$$\lambda_{1,2} = \frac{Tr \pm \sqrt{Tr^2 - 4\Delta}}{2}$$

$Tr = a - a = \emptyset$

$\Delta = -a^2 + b^2 = -(a^2 - b^2) = b^2 - a^2$

$\lambda_{1,2} = \pm \sqrt{-b^2 + a^2}$

- $a^2 > b^2$

$$\begin{pmatrix} a & b \\ -b & -a \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \sqrt{a^2 - b^2}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

a [redacted]

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$a, b \neq 0$

$$ax_1 + bx_2 = \sqrt{a^2 - b^2}\, x_1$$

$$\left(a - \sqrt{a^2 - b^2}\right) x_1 = -bx_2$$

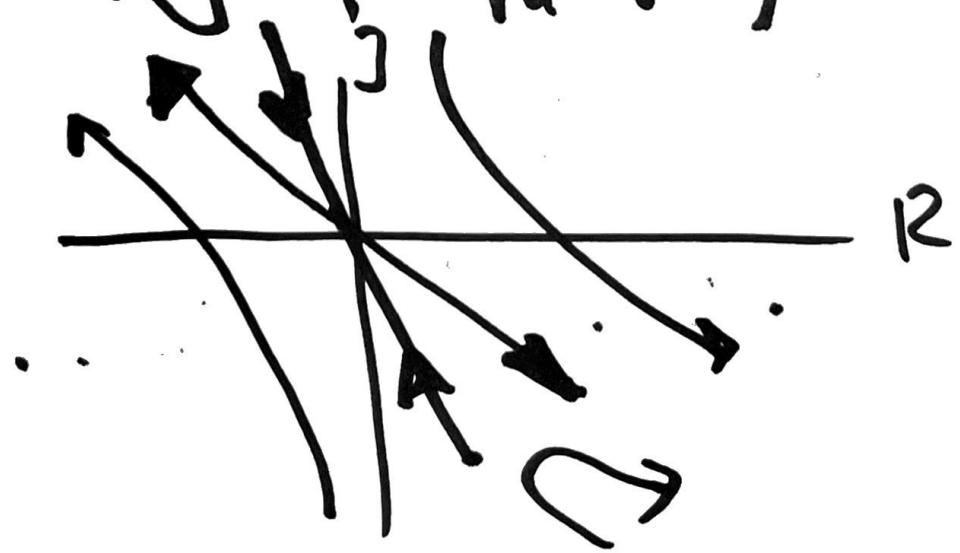
$$\underline{x}_1 = \begin{pmatrix} -b \\ a - \sqrt{a^2 - b^2} \end{pmatrix}$$

$$a - \sqrt{a^2 - b^2} > 0$$

- $$ax_1' + bx_2' = -\sqrt{a^2 - b^2}\, x_1'$$

$$\left(a + \sqrt{a^2 - b^2}\right) x_1' = -bx_2'$$

$$\underline{x}_1' = \begin{pmatrix} -b \\ a + \sqrt{a^2 - b^2} \end{pmatrix}$$



What happens when two identical cautious people meet if they are both responsive.

Two identical, cautious, responsive

$$\frac{d}{dt} R = a R + b J \qquad a < 0 \quad b > 0$$

$$\frac{d}{dt} J = b R + a J$$

$$A = \begin{pmatrix} a & b \\ b & a \end{pmatrix} \qquad T_r = 2a \qquad \Delta = a^2 - b^2$$

$$\lambda_{1,2} = a \pm \frac{1}{2}\sqrt{4a^2 - 4a^2 + 4b^2}$$

$$= a \pm |b| = a \pm b$$

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- $\begin{pmatrix} a & b \\ b & a \end{pmatrix}\begin{pmatrix} x_{11} \\ x_{12} \end{pmatrix} = (a+b)\begin{pmatrix} x_{11} \\ x_{12} \end{pmatrix}$

$a < 0$, $b > 0$

$$a x_{11} + b x_{12} = a x_{11} + b x_{11}$$

$$b x_{12} = b x_{11} \Rightarrow \underline{x}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$(a+b) = \lambda_1$; $\underline{x}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

- $\begin{pmatrix} a & b \\ b & a \end{pmatrix}\begin{pmatrix} x_{21} \\ x_{22} \end{pmatrix} = (a-b)\begin{pmatrix} x_{21} \\ x_{22} \end{pmatrix}$

$$a \underline{x}_{21} + b x_{22} = a \underline{x}_{21} - b x_{21}$$

$$x_{22} = -x_{21}$$

$(a-b) = \lambda_2$; $x_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

8

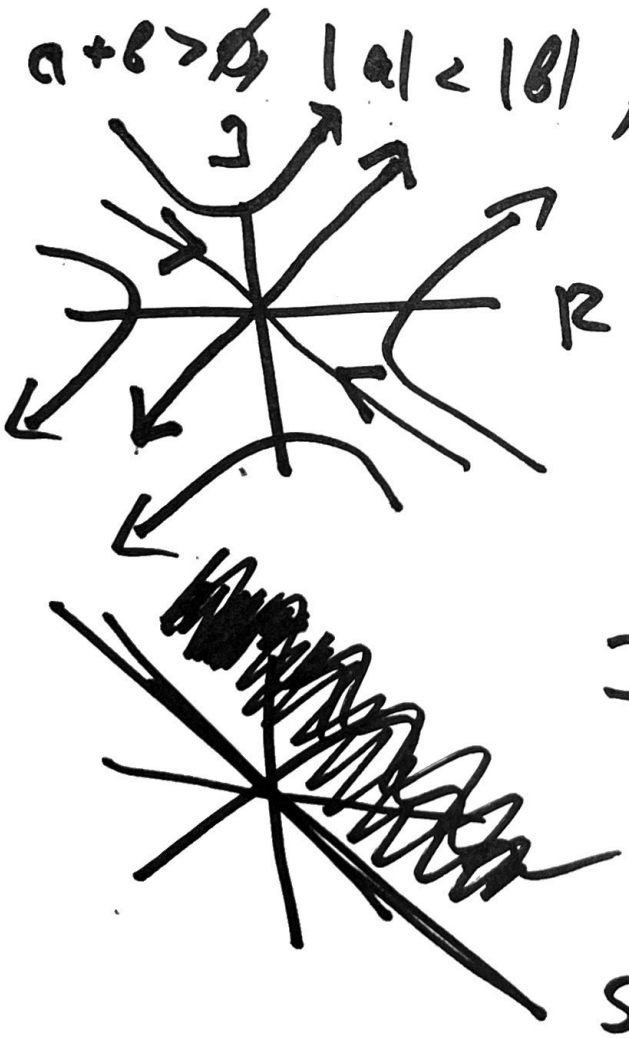
$\lambda_1 = (a+b)$; $\underline{x}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ $a < 0$, $b > 0$

$\lambda_2 = a-b$ $\underline{x}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

(a) $a+b > 0$, $|a| < |b|$

(b) $a+b < 0$, $|a| > |b|$

$a+b > 0$ $|a| < |b|$, more responsive than cautious



Love Love

Hate - Hate

$J > -R$

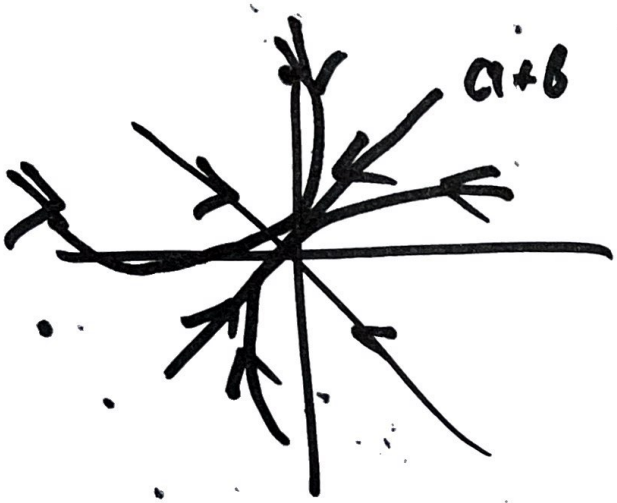
$\underline{J + R > 0}$

Set of initial conditions leading to LL

-9- $a+b < \emptyset \rightarrow \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ $|a+b| < |a-b|$

$a-b < \emptyset \rightarrow \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ more cautious then responsive

$a < \emptyset$
$b > \emptyset$





$$\frac{d}{dt}\underline{x}(t) = \underline{\underline{A}}\,\underline{x}(t)$$

$$\underline{x}(t) = \underline{\underline{x}}$$

$$= e^{\underline{\underline{A}}t}\,\underline{x}(t=0),$$

$\underline{\underline{A}} \in R^{2\times 2}$;

λ_1, λ_2 are both real and

$\lambda_1 > \lambda_2 > 0$ different

$\lambda_1 < 0$

$\lambda_2 > 0$

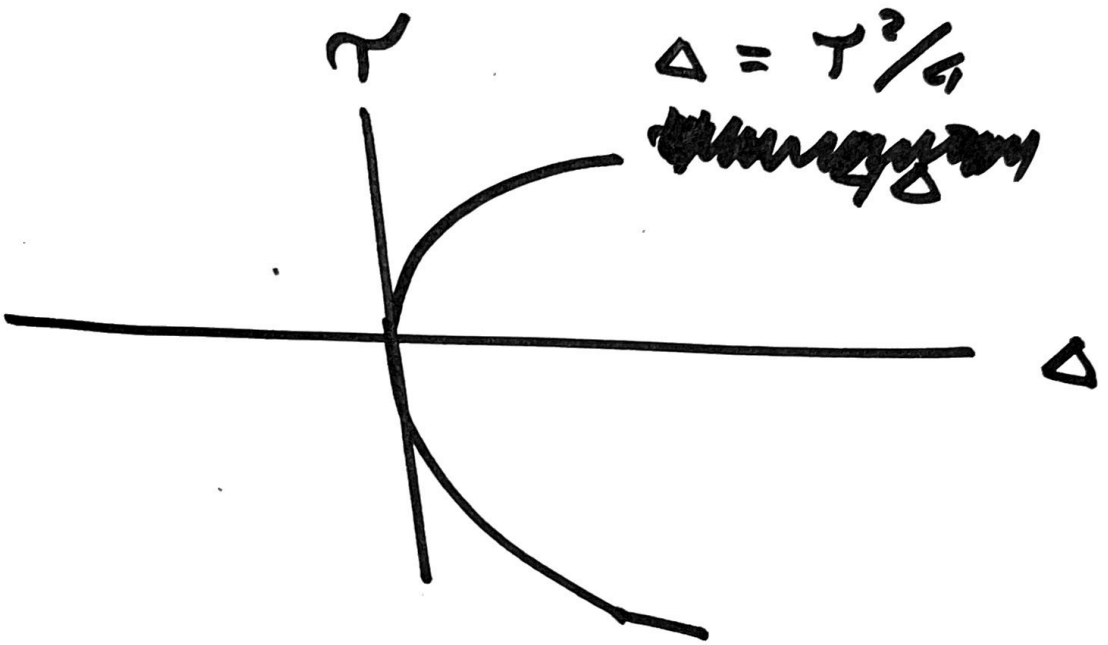
$\lambda_1 = \lambda_2$ → 2 eigenvectors

→ 1 eigenvector (deffective)

→ $\lambda_1 = \lambda_2^*$

2 complex roots that are cc of each other

$$\lambda_{1,2} = \frac{Tr \pm \sqrt{Tr^2 - 4Det}}{2}$$

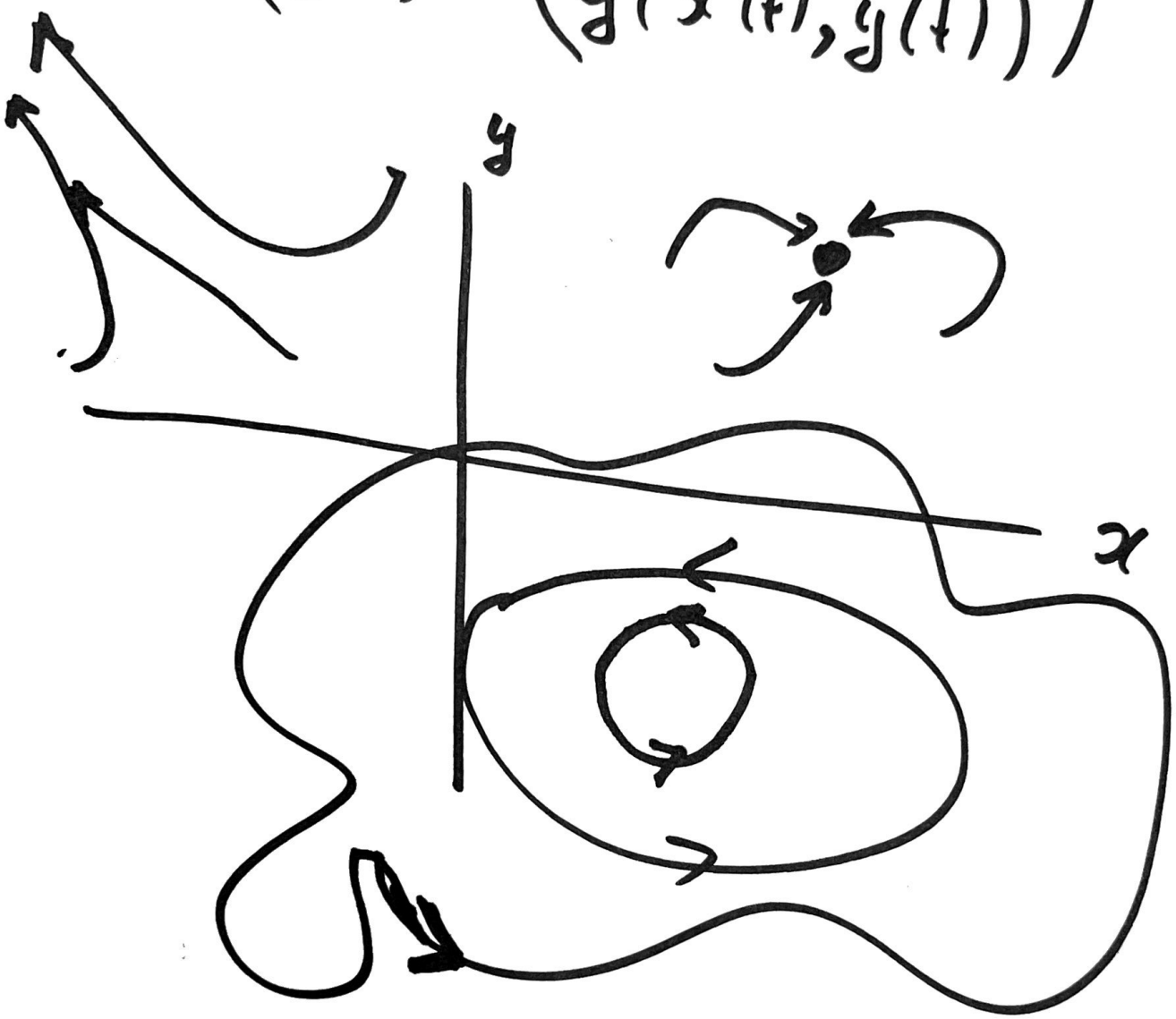


$$\underline{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$$

$$\underline{f}(\underline{x}(t)) = \begin{pmatrix} f_1(x_1\ x_2 \ldots x_n) \\ f_2(x_1\ x_2 \ldots x_n) \\ \vdots \\ f_n(x_1\ x_2 \ldots x_n) \end{pmatrix}$$

$$\frac{d}{dt}\underline{x}(t) = \underline{f}(\underline{x}(t))$$

$$\frac{d}{dt}\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} f(x(t), y(t)) \\ g(x(t), y(t)) \end{pmatrix}$$



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Fixed point

$$\underline{x}^* = (x^*, y^*)$$

$$f(x^*, y^*) = g(x^*, y^*) = 0$$

$$\dot{\underline{x}}(t) = f(\underline{x}(t))$$

IF

$\underline{x}(t) = \underline{x}^*$, such that

$$\underline{f}(\underline{x}^*) = 0$$

$\underline{x}(t) = x^* \quad \forall t$

$$\left.\begin{aligned}\dot{x} &= f(x,y)\\ \dot{y} &= g(x,y)\end{aligned}\right\} (1)$$

Let (x^*, y^*) be a fixed point $f(x^*, y^*) = g(x^*, y^*) = \emptyset$

$$(2)\begin{cases} x(t) = x^* + \eta(t) & |\eta| << 1 \\ y(t) = y^* + \nu(t) & |\nu| << 1 \end{cases}$$

(2) → (1)

$$\underbrace{\frac{d}{dt}x^*}_{=0} + \frac{d}{dt}\eta(t) = f(x^* + \eta(t), y^* + \nu(t))$$

$$\underbrace{\frac{d}{dt}y^*}_{=\emptyset} + \frac{d}{dt}\nu(t) = g(x^* + \eta(t), y^* + \nu(t))$$

$$\frac{d}{dt} z(t) = f(x^* + z, y^* + v) = f(x^*, y^*) + \frac{\partial f(x^*, y^*)}{\partial x} z + \frac{\partial f(x^*, y^*)}{\partial y} v$$

$$\frac{d}{dt} y(t) = g(x^* + z, y^* + v) =$$

$$= g(x^*, y^*) + \frac{\partial g(x^* y^*)}{\partial x} z + \frac{\partial g(x^* y^*)}{\partial y} v$$

$$\frac{d}{dt}\begin{pmatrix} z(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} \frac{\partial f(x^*,y^*)}{\partial x} & \frac{\partial f(x^*,y^*)}{\partial y} \\ \frac{\partial g(x^*,y^*)}{\partial x} & \frac{\partial g(x^*,y^*)}{\partial y} \end{pmatrix} \begin{pmatrix} z(t) \\ y(t) \end{pmatrix}$$

Consider initial value problem

$$\begin{cases} \frac{d}{dt}\underline{x}(t) = \underline{f}(\underline{x}(t)) \\ \underline{x}(t=t_0) = \underline{x}_0 \end{cases} \quad (*)$$



Let $\underline{f}(\underline{x})$ is continuous along with its first partial derivatives on an open connected set

Then solution to IVP $(*)$ exists and is unique in some interval

$$t \in [t_0 - \tau, t_0 + \tau]$$

$$\tau > 0$$